

Exercise 1:

Resistance based on the color code: 1 k Ω (1000 ohms)

Measured resistance using the Multimeter:

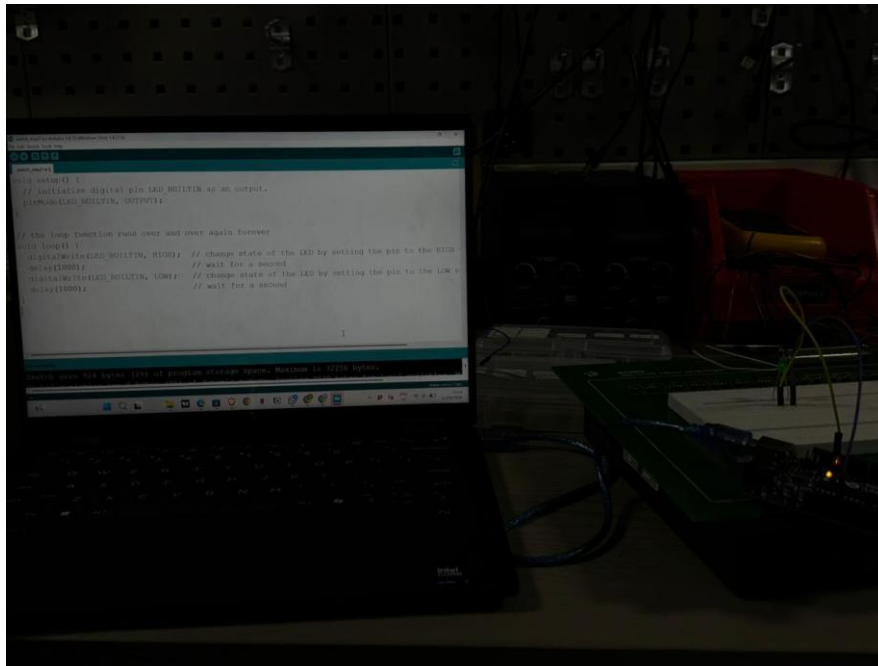


Experiment Reflection: Where does the difference in the resistance values arise from?

The discrepancy between the theoretical 1 k Ω value and our recorded 0.977 k Ω reading likely comes down to two reasons we came up with: multimeter accuracy and contact resistance. First, the digital multimeter is not a perfectly precise device; its internal circuitry has its own tolerance and margin of error, meaning the meter itself can slightly misread the exact value by a few ohms. Additionally, when you press the metal probes against the resistor's legs, that physical connection creates a tiny bit of extra contact resistance which we noticed because as we tightened and loosened our grip on the metal probes against the resistor's legs, the value would shift, therefore causing slight fluctuations in the final reading we saw on the screen.

Exercise 2:

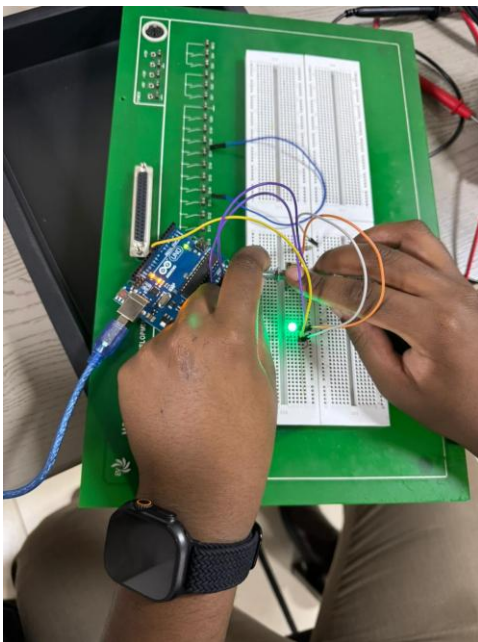
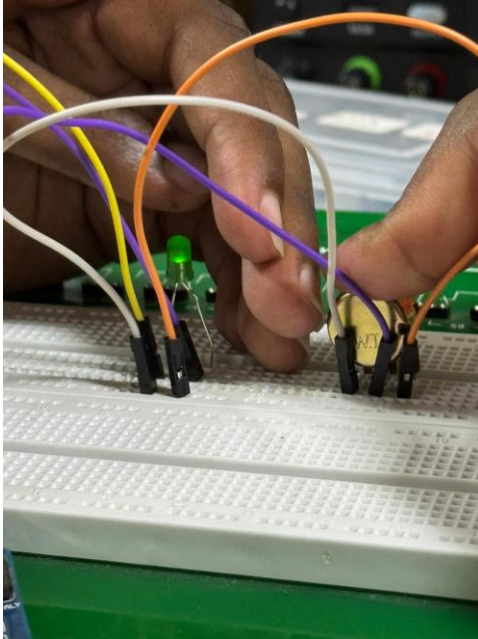
1. Physical implementation of the setup:

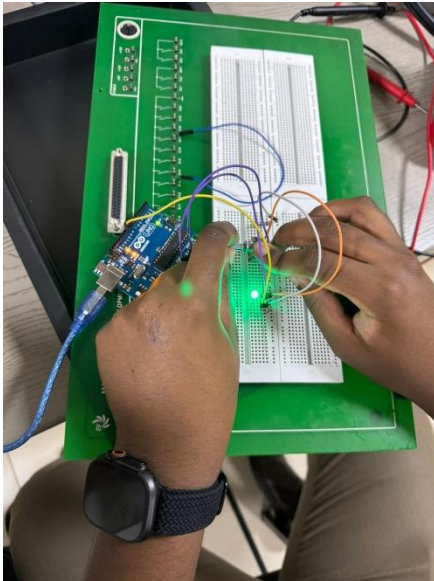


2. The code used in Arduino is also captured in the image above.
3. Voltage passing through the circuit per the multimeter.



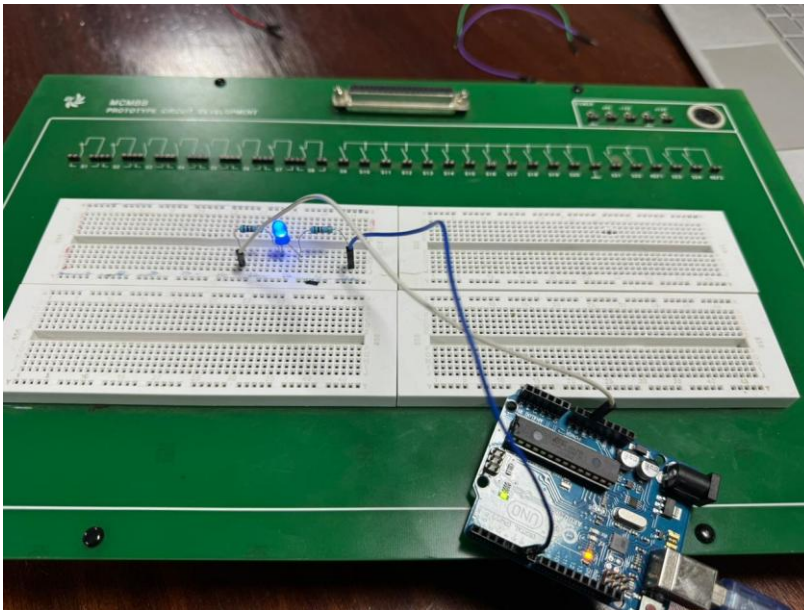
4. Three images of the modified circuit with the added potentiometer set at 25%, 50% and 100% brightness (respectively).



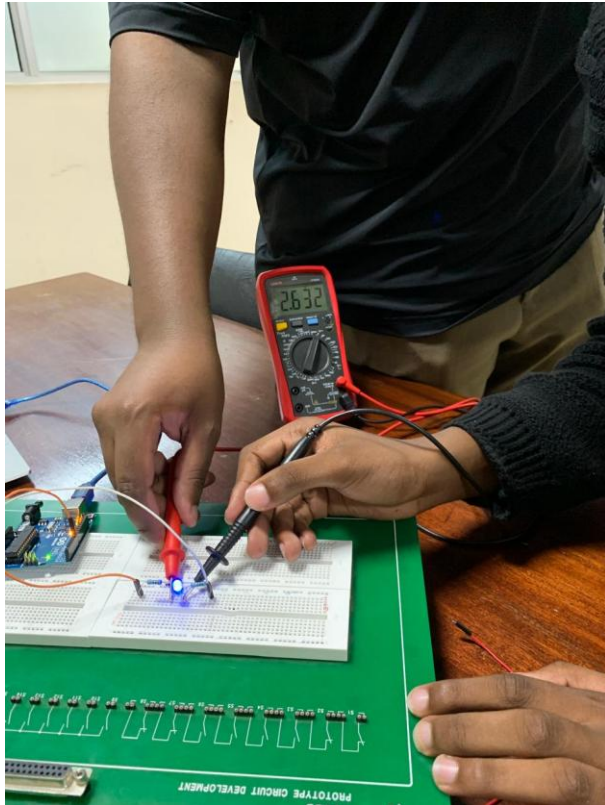


Exercise 3:

1. Physical implementation

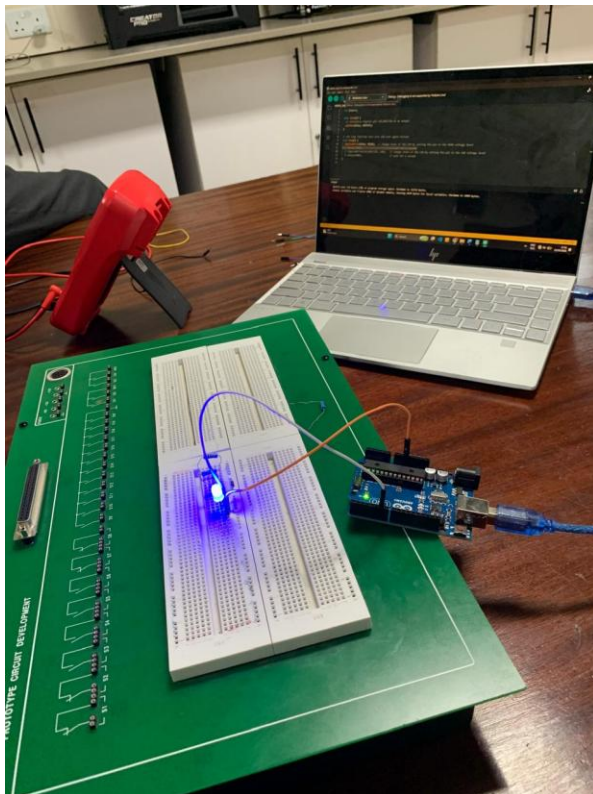


2. The voltage measured by the multimeter across the circuit.



Exercise 4:

1. Physical implementation:



2. The blink program code is also displayed in the image

3. Multimeter measurement of the voltage that passed through the resistors.

